

PART 1

Basic Concepts & Interview Questions

Preparation for SOC Level 1 (Covering network fundamentals)

Networking & Cybersecurity Essentials: Key Concepts Explained

Understanding networking and security fundamentals are crucial in today's digital world. Here's a breakdown of key topics with real-world applications:

◆ Switch

A switch is a networking device that facilitates communication between multiple devices within a Local Area Network (LAN).

✓ Operates at Layer 2 (Data Link Layer) of the OSI model ✓ Uses MAC addresses to forward data to the correct device ✓ Improves efficiency compared to hubs, which broadcast data to all devices

✓ Real-World Example: In a corporate office, a switch ensures that when an employee sends an email, data is directed only to the intended recipient instead of flooding the entire network—improving performance and efficiency.

◆ Router

A router forwards data between different networks, ensuring devices can communicate over the internet.

✓ Operates at Layer 3 (Network Layer) ✓ Uses IP addresses to determine the best route for data transmission ✓ Enables internet access, security features, and efficient routing

✓ Real-World Example: In an airport Wi-Fi network, thousands of passengers connect to the internet through a router system. The router efficiently manages traffic, ensuring secure connections and stable performance across multiple access points.

◆ OSI Model

A seven-layer framework that standardizes network communication:

- 1 Physical – Hardware components (cables, signals)
- 2 Data Link – MAC addresses, switches
- 3 Network – IP addressing, routing
- 4 Transport – TCP/UDP protocols for reliable transmission
- 5 Session – Manages session interactions between applications
- 6 Presentation – Translates, encrypts, compresses data
- 7 Application – Provides direct network services (web browsers, emails)

✓ Real-World Example: Sending a letter:

- The Physical layer represents paper and envelopes
- The Network layer defines postal system routing
- The Transport layer ensures reliable delivery
- The Application layer handles sender-receiver communication

◆ TCP/IP Model

The standard protocol suites for internet communication, divided into four layers:

- ✓ Link Layer – Handles physical connectivity (Ethernet, Wi-Fi)
- ✓ Internet Layer – Manages IP addressing and routing
- ✓ Transport Layer – Ensures reliable delivery via TCP/UDP
- ✓ Application Layer – Enables web browsing, email, file transfers

✓ Real-World Example: Like sending a package:

- The Link Layer is the delivery truck
- The Internet Layer ensures correct routing
- The Transport Layer guarantees safe delivery
- The Application Layer enables package use

◆ Cisco Proprietary Protocols

Exclusive networking protocols designed by Cisco:

✓ EIGRP – Efficient routing for enterprise networks

✓ HSRP – Ensures high availability for routers

✓ CDP – Helps Cisco devices share network information

✓ VTP – Simplifies VLAN management across Cisco switches

✓ PVST – Optimizes network traffic per VLAN

✓ Alternatives: For example, OSPF is an industry-standard alternative to EIGRP, and VRRP serves a similar purpose to HSRP but is vendor-neutral.

◆ STP (Spanning Tree Protocol)

Prevents network loops in switched networks by dynamically selecting the best data transmission path while blocking redundant links.

✓ Prevents broadcast storms & data duplication

✓ Elects a Root Bridge to optimize traffic flow

✓ Variants like RSTP provide faster convergence

✓ Real-World Example: STP functions like traffic lights at intersections—without them, network packets could collide in loops, causing congestion.

◆ TLS (Transport Layer Security)

A cryptographic protocol securing communication over the internet. Successor to SSL, TLS ensures data encryption, integrity, and authentication in web browsing, email, and secure transactions.

✓ Used for HTTPS (port 443), secure emails (IMAPS, SMTPS) ✓ Enhances security over older SSL versions ✓ Latest versions (TLS 1.2 & TLS 1.3) offer stronger encryption

✓ Real-World Example: E-commerce websites use TLS encryption to protect online payments, preventing hackers from stealing credit card details.

Cybersecurity Essentials: Encryption, Threats, SIEM

◆ Encryption & Decryption

✓ Encryption: Converts readable plaintext into unreadable ciphertext using cryptographic algorithms to ensure confidentiality.

✓ Decryption: Converts ciphertext back into plaintext using a secret key.

✓ Real-World Scenario: Online banking encrypts credit card details to prevent hackers from intercepting sensitive data.

📌 Interview Tip: "Encryption protects data confidentiality, while decryption restores it using a key."

◆ Hashing

✓ Converts data into a fixed-length hash value using a one-way function, ensuring integrity.

✓ Used in password storage, data integrity checks, and digital signatures.

✓ Real-World Scenario: Websites store hashed passwords so that even if attackers steal the database, they cannot reverse the hash to retrieve the actual password.

📌 Interview Tip: "Hashing is irreversible and ensures integrity. Encryption can be reversed with a key and ensures confidentiality."

◆ Encoding & Decoding

✓ Encoding: Converts data into another format for compatibility, not security.

✓ Decoding: Converts encoded data back to its original format.

✓ Real-World Scenario: Base64 encoding is used in emails to ensure smooth transmission of attachments.

📌 Interview Tip: "Encoding is for compatibility; encryption is for security. Encoding can be reversed without a key."

◆ SIEM (Security Information and Event Management)

✓ Collects, analyzes, and correlates security logs to detect cyber threats.

🔥 Popular SIEM Tools: Splunk, IBM QRadar, ArcSight, Microsoft Sentinel, LogRhythm.

📌 Interview Tip: "SIEM provides real-time threat detection by correlating logs from different security sources."

◆ Cyber Threats Explained

✓ Phishing – Attackers trick users into revealing sensitive information using fake emails or websites.

✓ Malware – Includes viruses, worms, Trojans, ransomware, and spyware that harm systems or steal data.

✓ Privilege Escalation – Hackers gain higher-level access by exploiting vulnerabilities.

📌 Interview Tip: "Phishing is social engineering, while malware includes ransomware and Trojans that infiltrate systems."

◆ CIA Triad – Foundation of Cybersecurity

✓ Confidentiality – Protects sensitive data from unauthorized access.

✓ Integrity – Ensures data remains unaltered.

✓ Availability – Ensures data is accessible when needed.

📌 Interview Tip: "The CIA Triad defines cybersecurity—Confidentiality, Integrity, and Availability."

◆ Vulnerability, Threat & Risk Explained

✓ Vulnerability – A weakness that can be exploited (e.g., unpatched software).

✓ Threat – Potential danger exploiting a vulnerability (e.g., malware, insider attack).

✓ Risk – The likelihood of a threat successfully exploiting a vulnerability.

✅ Real-World Scenario:

- Leaving a door unlocked = Vulnerability
- A thief targeting the house = Threat
- Likelihood of break-in based on crime rate = Risk

📌 Interview Tip: "Risk assessment helps organizations prioritize cybersecurity investments."

◆ Steps After Detecting a Security Breach

- 1 Isolate affected systems
- 2 Analyze attack patterns
- 3 Identify the root cause
- 4 Apply patches & fixes
- 5 Report on the incident
- 6 Strengthen security measures

📌 Interview Tip: "Incident response requires immediate containment and forensic analysis to prevent further damage."

MITRE ATT&CK – Simple Explanation with Practical Use

MITRE ATT&CK is a framework that helps cybersecurity teams understand how hackers attack systems. It provides a detailed list of tactics and techniques used by cybercriminals, helping security analysts detect, investigate, and stop threats effectively.

- **Tactics:** What attackers want to achieve (e.g., Initial Access, Persistence).
- **Techniques:** How they do it (e.g., Phishing, Credential Dumping).
- **Procedures:** Real-world execution used by hacker groups.

◆ EDR vs. XDR – Endpoint & Extended Detection Response

Feature → EDR (Endpoint Detection & Response): XDR (Extended Detection & Response)

Scope → Monitors endpoints (laptops, servers): Covers endpoints, cloud, email & network

Detection → Identifies malware & exploits : Correlates security data across multiple layers

Response → Isolates infected endpoints : Coordinates actions across entire IT infrastructure

✅ Real-World Scenario:

- EDR: Detects ransomware on an employee's laptop & isolates the machine.
- XDR: Detects phishing email, correlates endpoint alerts, and prevents an attack.

📌 Interview Tip: "XDR integrates threat intelligence across multiple security layers."

◆ DLP (Data Loss Prevention)

✓ Monitors & prevents unauthorized access, sharing, or leakage of sensitive data.

✓ Used for protecting financial records, personal data, and intellectual property.

✓ Real-World Scenario: A healthcare provider blocks employees from accidentally emailing patient records externally using DLP rules.

📌 Interview Tip: "DLP protects confidential data by restricting unauthorized transfers."

Understanding the OWASP Top 10 vulnerabilities is non-negotiable in 2025.

What is OWASP and why is it important in 2025?

OWASP (Open Worldwide Application Security Project) is a nonprofit organization that provides free, community-driven resources like the OWASP Top 10, helping developers and businesses secure web applications against common vulnerabilities.

Here's a simplified breakdown I reviewed recently that makes it incredibly easy to grasp:

- 1 Broken Access Control – Users doing what they shouldn't be able to.
- 2 Cryptographic Failures – Weak encryption or unencrypted sensitive data.
- 3 Injection Attacks – Malicious input tricking apps into unsafe behavior.
- 4 Insecure Design – Flawed logic allowing abuse (e.g., reservation flooding).
- 5 Security Misconfiguration – Default settings, unnecessary services, sample scripts.
- 6 Vulnerable Components – Outdated libraries and exposed systems.
- 7 Auth Failures – Flaws in session, password, or token handling.
- 8 Integrity Failures – Untrusted software or insecure CI/CD pipelines.
- 9 Logging & Monitoring Failures – Lack of visibility into suspicious activity.
- 10 SSRF – Tricking servers into accessing internal services behind the firewall.

How to Use Wazuh in Daily Workflow – SOC Analyst Perspective

Wazuh is a free and open-source security platform used for Extended Detection and Response (XDR) and Security Information and Event Management (SIEM). It helps SOC analysts monitor, detect, and respond to security incidents across various environments, including on-premises, virtualized, containerized, and cloud-based infrastructures.

◆ Daily Workflow with Wazuh – SOC Analyst Use Cases

- ✓ 🔍 Investigating Alerts – Wazuh provides real-time security alerts triggered by anomalies or potential cyber threats. SOC analysts review these alerts, correlate event data, and determine if further action is needed.
- ✓ 📊 Monitoring Endpoint Activity – SOC teams use Wazuh to track behavior across endpoints (servers, workstations, cloud assets). This includes file integrity monitoring (FIM), process execution tracking, and malware detection to ensure security posture.
- ✓ ⚠️ Correlating Logs for Attacks – Wazuh collects logs from various sources (firewalls, intrusion detection systems, cloud environments), helping analysts correlate attack patterns and detect suspicious activities like privilege escalation or unauthorized access.
- ✓ ⌚ Triage & Escalation Decisions – SOC analysts prioritize alerts based on severity and impact, escalating incidents when critical vulnerabilities or breaches are detected. Wazuh helps in triage automation, reducing response time.
- ✓ 📌 Great Balance of Visibility + Control – By integrating with SIEM platforms (Splunk, Elastic Stack) and other security tools, Wazuh provides a comprehensive security view, giving SOC teams the ability to investigate, detect, and respond efficiently.

Choosing the right routing protocol is basically picking your network's personality

- 👤 OSPF (The Perfectionist): I know every connection in the room and I calculate the fastest way to talk to each person, let me check my link-state database first.
- 🚗 EIGRP (The Smooth Operator): I'm efficient, fast, and I don't waste time recalculating the entire network just the parts that need updating.
- 👴 RIP (The Old-Timer): I've been doing this since the 80s. Sure, I might take a few extra hops to reach my destination, but hey who needs speed when you have experience.

What are BGP and OSPF?

BGP (Border Gateway Protocol) and OSPF (Open Shortest Path First) are routing protocols used in computer networks.

BGP is an exterior gateway protocol used to exchange routing information between different autonomous systems.

OSPF is an interior gateway protocol used to determine the best path for data packets within an autonomous system.

BGP operates at the network layer and is commonly used in large-scale networks, such.

How does a packet flow over the internet

A packet flows over the internet by being encapsulated in various network layers and routed through different devices.

Packet is created at the application layer with data and destination IP address

Packet is then encapsulated in a frame at the data link layer with source and destination MAC addresses

Frame is transmitted over physical medium to the next hop based on routing table

At each hop, the frame is decapsulated and re-encapsulated based on the destination IP address

Packet reaches its destination after traversing multiple routers and switches

NETWORKING CHEAT SHEET			
OSI MODEL VS. TCP/IP MODEL • Application: Application • Presentation: Transport • Session: Internet • Transport: Network Access	TCP VS. UDP • Connection-oriented • Reliable • Error checking	DNS Domain Name System	
IP ADDRESSING IPv4 - 192.168.1.1 Private Ranges - 10.0.0.0/8 - 172.16.0.0/12 - 192.168.0.0/16	IPv6 Compress (::1) Link-Local Prefix fe80::a/10	DHCP Dynamic Host Configuration Protocol	
NETWORKING COMMANDS - ping: Check / route - traceroute: Display route - ipconfig /ifconfig: IP config, netconfig - netstat /s: Connection stats - nslookup: Arp queries	COMMON PORTS 20-21 FTP 22 SSH 23 Telnet 25 SMTP 53 DNS 80 HTTP 110 POP3 443 HTTPS 3389 RDP	NAT Network Address Translation	
ROUTING & SWITCHING - Router: IP forward between - Switch: Learns LAN, Learns MAC addresses	FIREWALLS, IDS, IPS, PROXY - Firewall: Packet Monitor - IDS: Prevention - IPS: Prevention - Proxy: learns for	VPN Virtual Private Network	
		NETWORK PROTOCOLS HTTP Hypertext Transfer Protocol	
		FTP File Transfer Protocol	
		ICMP Ping error messages	
		SNMP Network management	
		ARP Info. MAC address	
		RDS Keeps within a LAN	

TOP PHISHING EMAIL ANALYSIS TOOLS:

Top Phishing Email Analysis Tools

PhishTool  Combines threat intelligence, open-source intelligence (OSINT), and email metadata to provide a comprehensive phishing response platform.	ThePhish  An open-source tool that automates the analysis of phishing emails, extracting observables and providing verifiable threat intelligence based on integrated platforms like Threats, Cortex, and MISP.	PhishTank  A community-based platform where users submit and verify suspected phishing sites, offering a database of known phishing URLs.
OpenPhish  Provides real-time phishing intelligence by analyzing and verifying phishing URLs, aiding in the detection of phishing attempts.	Cuckoo Sandbox  An open-source automated malware analysis system that can process various malicious files and URLs, providing detailed reports on their behavior.	Any.Run  An interactive malware-fighting service that enables users to upload and watch the actions of suspicious files without running them on their own systems.
AbuseIPDB  A platform that provides a central repository for reporting and identifying IP addresses associated with malicious activity online.	PhishTitan  Developed by TitanHQ, this AI-driven solution offers advanced anti-phishing protection, particularly for Microsoft 365 users.	CheckPhish  Utilizes AI to detect phishing websites and emails, providing real-time analysis and threat intelligence.
Cyber Press		
IsttPhish  Offers phishing detection services by analyzing URLs and email content to identify potential threats.	OPSWAT Email Security  Provides advanced threat detection and prevention for email communications, including phishing analysis.	Email Veritas  A tool that verifies the authenticity of emails by analyzing headers and content to detect phishing attempts.
Microsoft SmartScreen  Integrated into Microsoft products, it analyzes emails for phishing and malware, providing warnings to users.	Proofpoint Email Protection  Offers comprehensive email security, including phishing detection and analysis, to protect organizations from email-based threats.	Barracuda Email Protection  Provides protection against phishing and business email compromise by analyzing message contents, formatting, and header information.
Avanan  Integrates with mail providers to protect against phishing by evaluating existing relationships between senders and recipients to establish a level of trust.	Fortinet FortiMail  Offers email security solutions that include phishing detection and analysis to protect against email-based threats.	Mimecast Email Security  Provides advanced phishing detection and analysis, leveraging AI and machine learning to protect against sophisticated email threats.